

CSCE 606 – Software Engineering
Project: Inventory System
Sprint # 4 – Planning

Customer: Dr. Robert Lightfoot

Development Team:

- Rafael Morales
- Khush Patel (Scrum Master)
- Saba Mostofi
- Arun Akash Rangaraj
- Sid Rayapur

Links:

- Deployed app: <https://robert-inventory-f91e26f91bb2.herokuapp.com/>
- Github Project: <https://github.com/orgs/tamu-edu-students/projects/101>
- Github Repo: <https://github.com/tamu-edu-students/CSCE-606-Inventory-System>
- Shared folder: <https://drive.google.com/drive/folders/1NsQfvHf9n272E8VKVghdkU4B7Elk4EYI>
- Discussion Channel: <https://app.slack.com/client/T08A4A8U89M/C08AJRT0UTC>

Sprint #4 Goal:

Our goal for this sprint is to significantly improve the mobile user experience by redesigning key web pages and interactions for responsiveness and usability. As well as, several critical bugs (specifically the ones affecting search, deletion, and pop-up behavior) will be taken care of and ensure smooth mobile operations. At the end of the sprint, the mobile interfaces (dashboard, items, bins, locations, and pop-ups) will be fully optimized, and the bugs will be resolved.

Sprint Backlog and User Stories:

Features:

- 1. Improve User Mobile Experience for Dashboard**
As a user of the website, I want to have all the components of the website visible on my mobile screen so that I can have a better experience.
Points: 2
Developer: Rafael
- 2. Improve Items Webpage for a Better User Mobile Experience**
As a user of the website, I want to have all items arranged in a friendly way on the mobile screen so that I can search, edit, update, and delete without any issues.
Points: 5
Developer: Arun
- 3. Improve Bins# (/bins/x) Show Webpage for a Better User Mobile Experience**
As a user, I want items within bins to be arranged in a mobile-friendly layout—including

pictures for each item—so that I can manage my bins easily.

Points: 3

Developer: Saba

4. Add Capabilities on “bin/x” for Remove, Add, and Edit Items Targeting User Mobile Experience

As a user, I want to be able to add, edit, and remove items directly from the bin page (/bins/x) using an interface optimized for mobile.

Points: 3

Developer: Saba

5. Improve Pop-up Windows for a Better User Mobile Experience

As a user, I want mobile-friendly pop-up windows (for LOG, SALE, and user profiles) so that I can manage my inventory effectively from my phone.

Points: 5

Developer: Khush

6. Create Logs Targeting User Mobile Experience

As a user, I want to create logs to track every movement in my inventory over a specific time range.

Points: 5

Developers: Khush / Sid

7. Create Garage Sales Targeting User Mobile Experience

As a user, I want to display items older than a specific time so that I can generate a report of items I would like to sell.

Points: 5

Developer: Rafael

8. Improve /locations Webpage for a Better User Mobile Experience

As a user, I want the /locations page to be mobile-friendly so that I can manage my inventory more effectively.

Points: 2

Developer: Arun

Assignment of Tasks:

Feature Tasks:

1. Improve User Mobile Experience for Dashboard (Rafael)

Tasks:

- Use Chrome DevTools (Inspect -> Mobile view) to review the current dashboard layout.
- Adjust CSS and layout to ensure full visibility and ease of use on mobile.

- Test across multiple screen sizes and update stylesheets accordingly.
- Develop and execute BDD/TDD tests for mobile compatibility.

2. *Improve Items Webpage for a Better User Mobile Experience (Arun)*

Tasks:

- Refactor the items page for mobile responsiveness.
- Convert item names into clickable links to the show page (/items/[id]).
- Add mobile-friendly edit/update/delete buttons on the show page.
- Implement a dropdown menu for bin selection in item creation/update forms.
- Update controllers, views, and database migrations as needed; create corresponding BDD/TDD tests.

3. *Improve Bins# (/bins/x) Show Webpage for a Better User Mobile Experience (Saba)*

Tasks:

- Redesign the bin details page to display items in a mobile-optimized layout.
- Include images for each item.
- Update controllers/views and test for touch interactions.
- Create BDD/TDD tests.

4. *Add Capabilities on “bin/x” for Remove, Add, and Edit Items (Saba)*

Tasks:

- Create a dedicated column on the bin page for CRUD action links.
- Re-assess and update icons to ensure they are clear and mobile-friendly.
- Integrate the functionality into the controllers and views, then test thoroughly with BDD/TDD.

5. *Improve Pop-up Windows for a Better User Mobile Experience (Khush)*

Tasks:

- Redesign pop-up windows (LOG, SALE, User profiles) to be responsive on mobile devices.
- Ensure pop-ups do not obstruct key content and are easy to dismiss.

- Update and test controllers/views and develop BDD/TDD tests.

6. *Create Logs Targeting User Mobile Experience (Khush / Sid)*

Tasks:

- Develop a mobile-friendly interface for creating logs, including optimized date range inputs.
- Update the database schema if required and integrate log creation into the controllers/views.
- Execute BDD/TDD tests.

7. *Create Garage Sales Targeting User Mobile Experience (Rafael)*

Tasks:

- Design a mobile interface to display items older than a specific threshold for generating garage sale reports.
- Update filtering logic in controllers/views and create new migrations if needed.
- Develop and run BDD/TDD tests.

8. *Improve /locations Webpage for a Better User Mobile Experience (Arun)*

Tasks:

- Enhance the /locations page layout for mobile responsiveness and intuitive navigation.
- Update search functionality and ensure the page renders well on small screens.
- Modify controllers/views as necessary and create BDD/TDD tests.

Bug Fix Tasks:

1. *Bin and Item Search Malfunction (Rafael)*

Tasks:

- Investigate the JavaScript and backend logic for the nav-tab search functionality.
- Fix event handler issues and verify the corrected behavior across devices.
- Create and execute BDD/TDD tests.

2. *Overlapping Pop-up Windows (Khush)*

Tasks:

- Analyze and adjust the pop-up trigger logic to prevent simultaneous activation.
- Implement fixes and test under various click scenarios.
- Develop and run BDD/TDD tests.

3. BIN Deletion Issue (Saba / Rafael)**Tasks:**

- Debug the deletion logic to ensure bins are properly removed and any contained items are marked as unassigned.
- Update the controller logic and database constraints as needed.
- Test extensively and create BDD/TDD tests.

4. ITEM Deletion Issue (Khush)**Tasks:**

- Review the deletion process for unassigned items and implement the necessary fix.
- Update controller logic and test deletion functionality across scenarios.
- Develop and run BDD/TDD tests.

5. LOCATION Deletion Constraint (Arun)**Tasks:**

- Enforce logic in the controller to prevent deletion of locations that contain bins.
- Provide user feedback on deletion attempts and update the UI accordingly.
- Test constraints and create BDD/TDD tests.